

An AI-Based Platform for Content Summarization and Idea Generation using a Research Lab’s Output

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Abstract—Publication lists expose outputs but do not directly support full-text discovery, evidence-linked question answering, or the formulation of bounded student projects. This paper reports an artefact-oriented engineering case study and controlled offline evaluation of a source-traceable research-intelligence platform for a laboratory corpus. The frozen catalogue contains 134 records and 98 locally available PDFs; title/content validation and extraction rules admit 96 papers and 719 page-aware chunks to the experimental indexes. The platform combines keyword search, a reproducible feature-hashing baseline, a pinned learned-dense encoder, hybrid retrieval, citation-returning extractive question answering, evidence-only and generated thesis-extension modes, controlled topic and author discovery, and append-only AI-review events. On a 50-case AI-reviewed silver set with creation and shuffled reinspection, held-out keyword retrieval achieved Recall@10 1.000 and MRR 0.947, while the development-tuned hybrid achieved 0.925 and 0.860; no Holm-corrected comparison established hybrid superiority. Claim review over 50 answers found a supported-claim rate of 0.995 but citation correctness of 0.625 and answer-point coverage of only 0.136. A 28-profile AI-proxy recommendation comparison produced an uncertain relevance difference of 0.024 (95% CI [-0.024, 0.071]). Eligible index coverage was 719/719, and citations were attached to 400/400 checkable claims; these traceability properties do not repair weak completeness or abstention and do not establish human advisory benefit. The contribution is a reproducible platform and evaluation protocol that separates traceability, empirical quality, and human validation.

Index Terms—research intelligence, scholarly search, retrieval-augmented generation, recommendation, source traceability, reproducibility

I. INTRODUCTION

Institutional publication archives make research visible, but a title-and-link list leaves several practical questions unanswered: which full-text passages address a topic, which evidence supports an answer, how a prior study might be extended, and what remains unverified. Scholarly search engines and graphs support document and expert discovery at large scale [1]–[4]; research-paper recommenders add content or profile matching [5], [6]; and retrieval-augmented generation (RAG) conditions generation on retrieved evidence [7]. Combining these capabilities in a small, reviewable laboratory system creates a different engineering problem: the system must preserve provenance and uncertainty even when metadata, PDFs, models, and human review are incomplete.

The TTLAB precursor collected and displayed publications, generated lay summaries and notifications, and supported a

manual podcast workflow; it described RAG as future work rather than an evaluated component [8]. The present system extends that workflow to full-text ingestion, page-aware retrieval, question answering, thesis-extension suggestions, topic/author exploration, and administrative review. We define *source-traceable* narrowly: a returned claim or suggestion retains persistent paper, chunk, page, section, provider, timestamp, warning, and review-state locators. Traceability enables checking; it does not imply entailment, factual correctness, novelty, feasibility, or endorsement.

The study asks:

- **RQ1:** Can a heterogeneous laboratory archive be converted into an internally consistent, provenance-preserving experimental corpus and complete indexes?
- **RQ2:** How do keyword, feature-hashing, learned-dense, heuristic-hybrid, and development-tuned hybrid retrieval compare on source-derived silver judgments?
- **RQ3:** To what extent are extractive answers supported, correctly cited, complete, and able to abstain?
- **RQ4:** Does the full Thesis Extension Finder improve an evidence-only baseline under an AI-assisted proxy review, what is held-out topic-label performance, and what author-identity/integrity limits remain?
- **RQ5:** To what extent is the bounded artefact reproducible and reviewable on documented local hardware, externally format-compatible, and governed by privacy-minimizing, authenticated, accessible evidence interfaces without re-distributing restricted PDFs?

The contributions are: (1) an authoritative corpus/index manifest that fails loudly on partial or stale derived state; (2) a modular platform that keeps source facts, paper-stated future work, inferred gaps, and system suggestions distinct; (3) executed retrieval, claim/citation, recommendation, topic, section, performance, and generated-output evaluations with raw evidence; and (4) a sanitized, hash-verified reproduction and release workflow. No recruited human participants were used, so usefulness and supervisor suitability remain outside the evidence.

II. RELATED WORK

A. Scholarly Discovery and Recommendation

Early digital-library systems such as CiteSeer automated scholarly-document acquisition and indexing [1]. ArnetMiner

and later scholarly graphs linked documents, authors, and topics for expert and literature discovery [2], [3]; OpenAlex exposes a broad open catalogue of scholarly entities and relationships [4]. S2ORC shows the value and difficulty of large structured full-text corpora [9]. These systems motivate discovery and identity functions, but a laboratory-scale deployment also needs explicit review state, source permissions, and failure diagnostics.

Research-paper recommendation commonly combines content, citation, and user signals, while evaluation is sensitive to incomplete relevance judgments and unrealistic protocols [5], [6]. Expert finding similarly infers candidate expertise from document evidence rather than availability or endorsement [10], [11]. Accordingly, this platform labels author-topic links as publication-derived evidence and does not present them as assignment decisions.

Table I positions the implemented platform against representative scholarly-discovery systems using only capabilities supported by their cited descriptions. The comparison is intentionally functional rather than a claim of quality superiority.

B. Retrieval, RAG, and Evaluation

BM25-style lexical ranking remains a strong information-retrieval baseline [12]. Feature hashing provides a memory-bounded sparse representation but is not a learned semantic encoder [13]. Sentence-BERT, DPR, and SPECTER demonstrate learned representations for sentence, passage, and scholarly-document retrieval [14]–[16]; BEIR shows that relative performance varies materially across tasks and domains [17]. This motivates comparison rather than assuming dense or hybrid superiority.

RAG retrieves evidence before generation [7]; variants investigate retrieval/generation coupling and self-reflection [18], [19]. Returned citations, however, do not alone establish claim support. RAGAS and ARES operationalize automated RAG evaluation, and FActScore decomposes generated text into atomic factual units [20]–[22]. We adapt atomic-claim inspection but identify all judgments as AI-assisted silver evidence rather than human ground truth.

C. Topic Models and Human Review

Probabilistic LDA and embedding-assisted BERTopic offer corpus-induced topics, but topic coherence does not by itself guarantee semantic interpretability [23]–[25]. Interactive machine-learning guidance further argues that people must be able to inspect, correct, and understand model behavior throughout the workflow [26]. This project therefore retains an inspectable controlled vocabulary, compares it with a pinned dense prototype, and records attributed review transitions rather than presenting automatically assigned labels as final truth.

III. METHODOLOGY

A. Design and Units of Analysis

The work is an artefact-oriented engineering case study with design-science characteristics plus controlled offline eval-

uation. The artefact is the repository, database schema, indexes, API, React interface, review log, experiment code, and documentation. Units of analysis are papers and chunks for corpus integrity; queries for retrieval; atomic claims, citation links, and answer points for QA; synthetic student profiles and ranked paper sets for recommendation; papers for topic classification; and pipeline stages for performance. The study reports software/document experiments and AI-assisted review only. TTLAB authorization to undertake the project and use the laboratory corpus is recorded as a project fact; it is not presented as research-ethics approval or blanket third-party redistribution permission.

B. Corpus Freeze and Inclusion

The frozen snapshot `corpus-04a010207327069a` begins with 134 catalogue records and 98 locally available PDFs. A paper enters the experimental corpus only when its metadata record, permitted local PDF, extracted text, and title/content checks are consistent. 36 records remain unavailable or need review, and 2 PDFs are excluded because their contents do not match the catalogue titles. The resulting 96 eligible papers produce 719 eligible chunks from 735 raw chunks. Each chunk stores page range, inferred section, text hash, and source hash. Unknown sections are retained rather than forced.

Keyword FTS and vector manifests are generated over the same ordered eligible-chunk list. Every manifest records the corpus snapshot/hash, configuration hash, code commit, model/revision, dimensions, normalization, index hash, ordered chunk hashes, and completeness. Atomic replacement precedes transactional status updates; a bounded demonstration index is isolated and cannot claim authoritative coverage. Startup, API diagnostics, and experiments reject count, hash, model, or status mismatches.

C. Silver Evaluation Procedure

The retrieval set contains 50 source-derived questions stratified across factual lookup, methods, results, limitations/future work, bibliographic lookup, topics/applications, comparison, broad intent, multi-paper synthesis, hard negatives, and unanswerable cases. One AI process created evidence-linked judgments; a later shuffled pass re-inspected source pages without system rankings. The 30/20 development/test split was fixed before tuning. This is annotation-consistency checking, not inter-rater agreement or human gold labeling.

For query q , judged relevant papers are G_q and the first k predicted papers are $P_{q,k}$. Set recall is

$$R_q@k = \frac{|G_q \cap P_{q,k}|}{|P_{q,k}|}, \quad (1)$$

while $\text{Hit}@k$ is the indicator that the intersection is non-empty. We also report $\text{Precision}@k$, reciprocal rank, and $\text{nDCG}@10$ for graded judgments. Aggregates use answerable queries; abstention and false-positive rates use unanswerable queries. 10000 query-bootstrap resamples provide percentile 95% intervals. Two-sided paired randomization tests compare

TABLE I
CAPABILITY COMPARISON BASED ON THE CITED SYSTEM DESCRIPTIONS. “REVIEW” MEANS AN EXPLICIT GENERATED-OUTPUT CORRECTION WORKFLOW, NOT ORDINARY CURATION; “–” MEANS NOT CENTRAL TO THE CITED DESCRIPTION, NOT ABSENCE.

System	Primary role	Full text	Page locator	Author–work links	Project advice	Output review	Offline quality eval.	Public-facing UI/API
CiteSeer [1]	autonomous scholarly indexing	Y	–	–	–	–	–	Y
ArnetMiner [2]	literature and expert mining	–	–	Y	–	–	Y	Y
OpenAlex [4]	open scholarly entity graph	–	–	Y	–	–	–	Y
TTLAB precursor [8]	publication display and outreach	N	N	Y	–	–	summary	Y
This work	laboratory research intelligence	Y	Y	Y ^a	Y ^b	AI ^c	Y	Y

^aPublication-derived only; no availability or endorsement claim. ^bGenerated suggestions are separated from paper evidence and are not supervisor-approved. ^cAI review was executed; the implemented human-review path has no human approval evidence.

the tuned hybrid against each baseline, with Holm-Bonferroni family correction.

The QA set reuses the same 50 questions. It evaluates the offline extractive sentence-overlap answerer with the current untuned heuristic-hybrid retriever; it must not be read as an evaluation of the best keyword or development-tuned retriever. Answers were segmented into atomic checkable claims, connected to returned citations and AI-reviewed silver reference answer points, then reviewed for support, citation correctness/completeness, answer-point coverage, unsupported general knowledge, and answerability. Recommendation evaluation uses 28 synthetic profiles covering skills, interests, time, data, difficulty, project type, and avoid-topics. The evidence-only arm returns ranked papers/passages; the full arm additionally constructs an explicitly labeled extension. Topic evaluation uses a 60-case multi-label silver set, including all 39 controlled labels and other/unknown cases, with 36 development and 24 held-out cases. Generated-output review inspects every 27 historical answers, 7 recommendation runs, and 14 paper artefacts, appending attributed hash-chained events.

IV. SYSTEM DESIGN AND IMPLEMENTATION

Figure 1 separates presentation, API/security, intelligence, and evidence state. The React/Vite frontend provides deep-linkable Dashboard, Papers, Search, Ask TTLAB, Thesis Extension Finder, Topic/Author Explorer, Evaluation, and Admin routes. It exposes index freshness, source snippets, generation provider/time, warnings, review state/type, and evidence-only alternatives. Profile inputs remain in browser/request memory unless a user explicitly invokes a persistence-capable workflow; the evaluated finder calls are non-persistent.

FastAPI exposes public read-only discovery and bounded generation endpoints. Reviewer/admin changes require an environment-configured actor and authenticated token. There is no committed default secret. The security boundary applies exact origin/host configuration, request and upload limits, rate controls, path and URL validation, PDF-bomb safeguards, redacted logging, protected review transitions, and append-only attributed audit events. Local-demo mode is explicit and is not described as production authentication.

PyMuPDF extraction records per-page diagnostics. Blank/scanned/mixed detection and an optional OCR adapter preserve page method and review state; the current frozen

corpus used no OCR, so OCR quality is supported only by engineering fixtures. The section detector uses heading/layout evidence and preserves Unknown; a 40-case AI-reviewed silver sample measured 0.900 exact accuracy, while 199/719 eligible chunks remain unknown.

Retrieval modes share eligibility and filter controls. Keyword ranking uses SQLite FTS. The 256-dimensional signed feature-hashing path is retained as a lexical-feature baseline. Learned dense retrieval uses the 384-dimensional all-MiniLM-L6-v2 sentence transformer at the revision pinned in the released index manifest, with L2 normalization and mean-pooled overlapping windows for long chunks. Model acquisition is explicit; an unavailable dense model produces a visible fallback rather than silent substitution. The heuristic hybrid uses pre-existing keyword/vector weights and 6 hand-authored expansion/boost/penalty terms. A separate development search selected keyword/dense weights 0.10/0.80 without consulting the held-out test set.

As shown in Fig. 2, the answerer returns extracted sentences plus inline chunk identifiers and structured citations. The runtime statuses `grounded`, `partial`, and `unsupported` summarize locator presence and weak lexical-overlap checks; they do not establish factual entailment. Optional Ollama integration was probed, but the installed client could not reach a local service; no model digest or comparison was invented. The finder first constructs a paper-fact record (work performed, explicit future work, inferred gap, or no gap found), then creates a separately labeled suggestion with MVP, stretch scope, skills, data needs, risk, and evaluation plan. Topic/author views use canonical author identities and describe expertise only as indexed-publication evidence.

V. RESULTS

A. RQ1: Corpus and Index Integrity

Table II reports inventory rather than effectiveness. The three authoritative indexes each cover 719/719 eligible chunks; raw/excluded chunks cannot enter evaluation. Twelve navigation-text author defects were removed, punctuation-only aliases were repaired idempotently, and unresolved possible same-person variants remain separate for external confirmation. DOI, abstract, and keyword fields remain empty when no verified source supplies them.

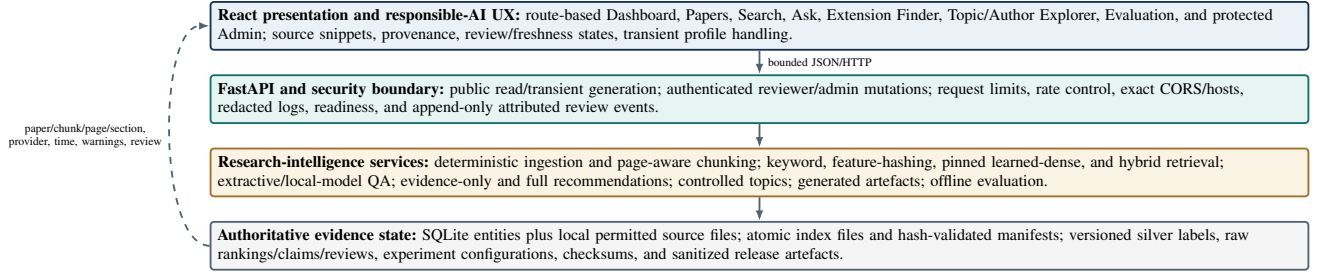


Fig. 1. Implemented architecture and evidence boundary. Source locators flow back through every layer; administrative mutations cross an authenticated boundary.

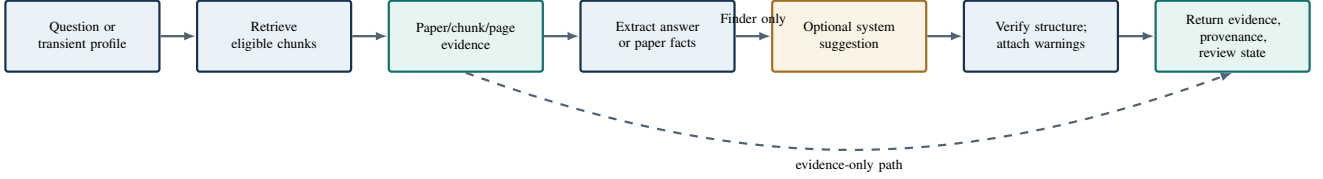


Fig. 2. Unified Ask/Finder workflow. The dashed evidence-only path and separate suggestion stage prevent generated advice from being attributed to a source paper.

TABLE II
FROZEN CORPUS AND AUTHORITATIVE DERIVED STATE.

Catalogue records	134	Local PDFs	98
Eligible papers	96	No-text/review	36
Raw chunks	735	Eligible chunks	719
Excluded mismatches	2	Unknown sections	199
Keyword coverage	719/719	Hashing coverage	719/719
Dense coverage	719/719	Dense dimensions	384

TABLE III
HELD-OUT RETRIEVAL ESTIMATES (19 ANSWERABLE QUERIES).

Mode	R@3	R@10	MRR	nDCG@10
Keyword	0.939	1.000	0.947	0.958
Feature hashing	0.654	0.704	0.588	0.582
Learned dense	0.916	0.989	0.939	0.939
Heuristic hybrid	0.795	0.917	0.813	0.829
Dev-tuned hybrid	0.856	0.925	0.860	0.871

B. RQ2: Retrieval Baselines, Tuning, and Ablation

Held-out estimates appear in Table III. Keyword retrieval had the highest point estimate on all four displayed measures: Recall@10 1.000 (95% CI [1.000, 1.000]), MRR 0.947 ([0.868, 1.000]), and nDCG@10 0.958 ([0.900, 0.998]). Learned-dense MRR was 0.939 ([0.842, 1.000]), and feature-hashing MRR was 0.588. The dev-tuned hybrid did not exceed keyword or dense. No Holm-corrected paired comparison rejected its null hypothesis, and no equivalence or non-inferiority test was specified; this study therefore provides no evidence of hybrid superiority or method equivalence.

Ablations expose why aggregate labels are insufficient. Removing only the topic adjustment raised tuned-hybrid Recall@10 from 0.925 to 0.978 (MRR 0.886), and removing all 6 heuristic terms recovered Recall@10 1.000 and MRR

TABLE IV
AI-ASSISTED CLAIM/CITATION REVIEW (CASE-CLUSTER BOOTSTRAP 95% CI).

Metric	Estimate	95% CI
Supported-claim rate	0.995	[0.987, 1.000]
Citation correctness	0.625	[0.564, 0.691]
Citation completeness	1.000	[1.000, 1.000]
Answer-point coverage	0.136	[0.067, 0.211]
Unsupported-claim rate	0.000	[0.000, 0.000]
Unanswerable abstention	0.000	[0.000, 0.000]

0.947, matching keyword-like behavior. Error coding identified 2 intent/ranking signals, 1 extraction/unknown-section signal, and 1 corpus-absence signal. Every mode returned results for the 1 held-out unanswerable case, yielding an unstable false-positive rate of 1.000; the 4-case QA unanswerable subset confirms that abstention is a real weakness.

C. RQ3: Claim and Citation Quality

The 50 offline answers contained 400 checkable claims. Table IV separates dimensions that a single “grounded” badge would conflate. Although strict supported-claim rate was 0.995 ([0.987, 1.000]) and every checkable claim carried a citation, only 0.625 ([0.564, 0.691]) of claim-citation links were fully correct. Answer-point coverage was 0.136 ([0.067, 0.211]), and none of 4 unanswerable cases abstained. Most errors were not hallucinated statements; they were supported but off-topic extracts, within-paper evidence misses, incomplete answers, and false-positive responses. Thus, high extractive support coexists with poor answer completeness and citation targeting.

D. RQ4: Recommendations, Topics, Authors, and Review

The evidence-only recommendation arm averaged relevance 0.631, compared with 0.655 for the full finder. The paired

difference 0.024 had CI [-0.024, 0.071], spanning zero; no benefit is demonstrated. Both arms passed all 84 source-fidelity checks; all 84 full-finder items also passed the fact/future-work/gap/suggestion separation check, which does not apply to the evidence-only arm. All 84 full-finder feasibility judgments were only partial, and 12/84 evaluation plans were partial. Sensitivity covered 21 perturbations plus the default (22 configurations) and found interest weighting most disruptive. These are AI-proxy judgments on synthetic profiles, not student or supervisor validation.

On 24 held-out topic cases, the controlled lexical method achieved precision/recall/F1 0.652/0.484/0.556 with 0.875 coverage. A pinned dense prototype traded precision for recall (0.417/0.694/0.521, coverage 0.958). The lexical vocabulary was retained for higher precision and inspectability, not because the experiment establishes universal superiority. Author auditing found no excluded-paper leakage, alias collision, authorship mismatch, or unsupported availability claim, while 13 possible same-person pairs remain external identity decisions.

The generated-output pass appended 48 hash-chained `codex-ai-review` events. 7 recommendation runs and 14 paper artefacts were regenerated against current chunks and marked `ai_reviewed`. The 27 historical RAG answers lacked enough stored generation scope/configuration for safe reproduction and were preserved but marked `needs_reprocess`. This is a correction workflow, not human approval.

Figure 3 shows the executed evaluation state rather than legacy placeholder files. Retrieval mode, sample boundaries, negative results, and dataset-specific reviewer or human-validation limits remain visible together. The Ask and Finder routes reuse the same provenance primitives for paper/chunk/page evidence.

E. RQ5: Performance and Reproduction

The full profile measured 17 stages with 3 process-cold and warm repetitions at concurrency one on AMD Ryzen 7 5800X 8-Core Processor (16 logical CPUs, 3.8 GiB RAM). The artifact is bound to source commit `5ccf22ec39cb` and provenance digest `61378b9aa78a`. Cold means a fresh Python process; operating-system caches were not flushed. Across the reported samples, 0 executions failed. Table V reports elapsed wall time and process peak RSS; it is a single-user characterization, not a saturation or capacity test.

VI. DISCUSSION

The principal engineering result is not a new retrieval algorithm but an evidence boundary that remains intact across ingestion, ranking, generation, review, API serialization, and user interface. RQ1 is answered positively for the frozen eligible subset: 719 chunks have one-to-one coverage in all authoritative indexes, and stale/partial indexes fail validation. This does not make the 36 unavailable or review-required records searchable or correct missing metadata.

RQ2 yields a useful negative result. Learned-dense MRR was 0.939 versus keyword MRR 0.947; equivalence was not

tested. Both heuristic and tuned hybrids had lower point estimates than the keyword baseline on this silver set, and the corrected paired analysis found no significant advantage. The result cautions against using the word “hybrid” as a quality claim and against treating feature hashing as semantic retrieval. The ablation pattern suggests that hand-authored topical adjustments can damage ranking even when they appear intuitively helpful.

RQ3 distinguishes faithfulness from usefulness. Extractive answers rarely invented unsupported material, yet often selected the wrong supported sentences, attached only partially correct citations, omitted answer points, and failed to abstain. Structural source traceability therefore makes failure inspectable but does not resolve it. A future answerer should use explicit answerability thresholds, retrieval calibrated to question intent, within-paper evidence selection, and claim-to-source verification before presentation.

RQ4 shows that recommendation structure can be auditable without being validated advice. All 84 AI-proxy-reviewed full-finder outputs kept evidence and suggestion fields separate, but the relevance difference is uncertain and feasibility remains unconfirmed. Likewise, controlled topics and publication-derived author links are implemented navigation features that may aid discovery; they are not proof of complete expertise. Human student, supervisor, author, and accessibility studies are deliberately left as future ethically governed work.

RQ5 is positive only within the authorized local boundary. The 17-stage, three-cold/three-warm, concurrency-one profile completed all 102 samples without failure on the documented WSL2 host; it is not load or capacity evidence. The clean-commit full reproduction and sanitized release verifiers passed while excluding PDFs, the live database, derived text, interface screenshots, secrets, and model files. The external harness mapped three CC BY JATS/XML articles into the production chunker contract and achieved 3/3 fixed lexical top-one matches; it did not test PDF ingestion or cross-domain quality. Public questions/profiles are transient and mutations authenticated. Automated accessibility regressions passed, but WCAG conformance and assistive-technology usability were not established.

VII. THREATS, ETHICS, AND RESPONSIBLE DEPLOYMENT

Construct validity: silver relevance, AI claim judgments, and proxy recommendation ratings may not measure human usefulness, trust, novelty, or feasibility. Citation correctness is claim-to-evidence support, not truth beyond the source. Performance and test counts are engineering properties, not effectiveness metrics.

Internal validity: within each two-pass silver dataset, the same AI procedure created and re-inspected labels without independent human adjudication, so correlated errors can survive both passes. Small held-out subsets create wide or degenerate intervals, especially the single retrieval unanswerable. Post-selection ablations are descriptive. The offline extractive answerer avoids stochastic model variance but is not representative of all RAG generators.

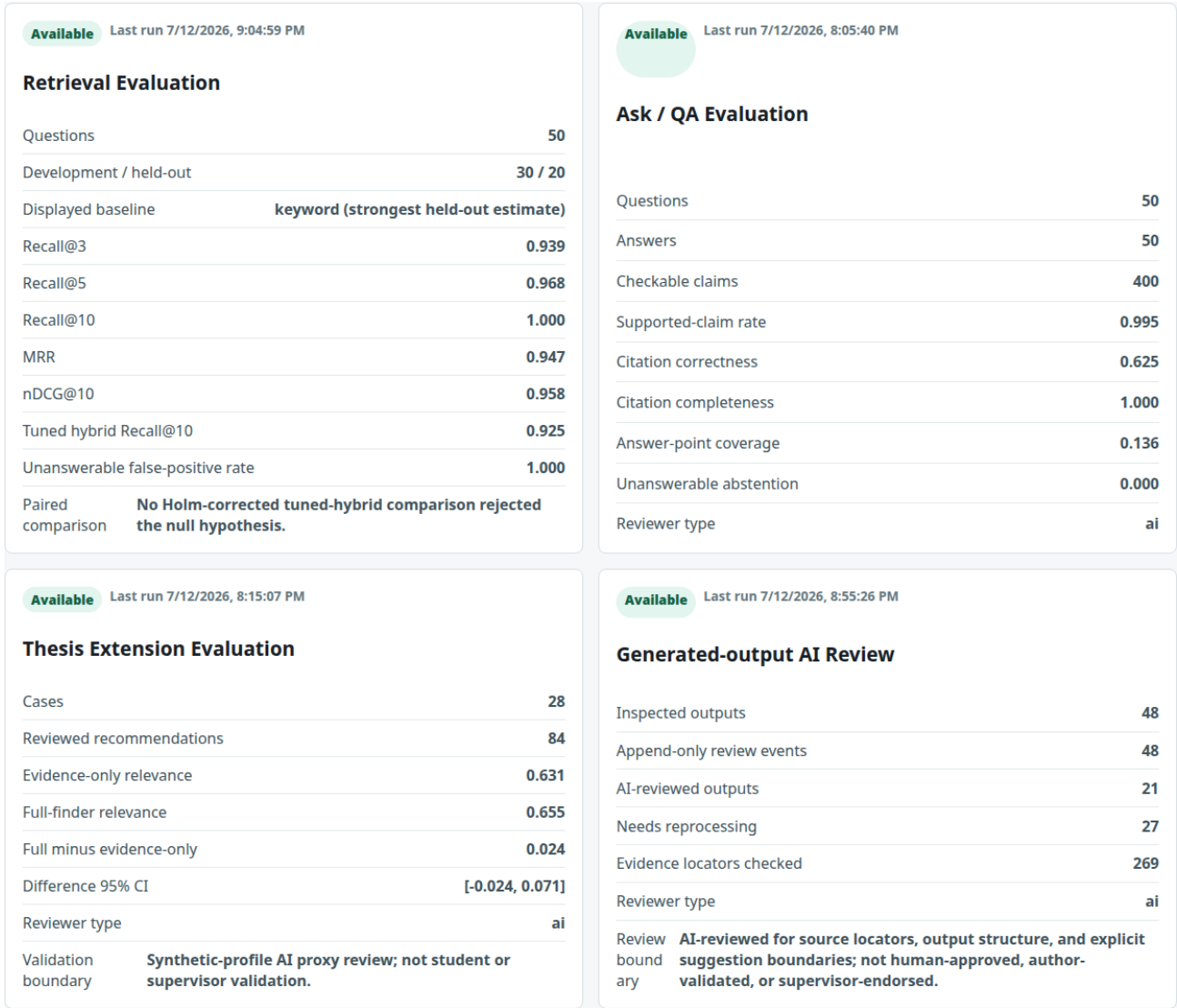


Fig. 3. Current Evaluation Dashboard over the frozen corpus. The interface separates retrieval, claim/citation, recommendation-proxy, and generated-output review results; AI-review labels are shown on the applicable evaluation cards.

External validity: results concern one laboratory snapshot and a small external format sanity check, not scholarly search generally. Thirty-six records have no eligible text and two mismatched PDFs were excluded. Topic vocabulary, author identities, English-language extraction, and local hardware further limit transfer.

Reproducibility validity: pinned inputs, revisions, seeds, raw rankings, manifests, and hashes reduce drift, but network discovery and operating-system caches remain variable. The learned model must be acquired explicitly, and venue PDF validation requires external credentials.

TTLAB explicitly authorized the project and use of the laboratory corpus. No participants were recruited, and no ethics approval or exemption is claimed. Authorization does not confer rights to redistribute every third-party PDF; source PDFs, the live database, credentials, and private backups are excluded from the sanitized bundle. Student profile inputs are

minimized and non-persistent by default. Public deployment requires authenticated review, controlled origins/hosts, rate limiting, corrections, backups, monitoring, and accessible interaction. Datasheet, model-card, and NIST AI risk-management practices inform the disclosure and governance approach [27]–[29]. Codex/LLM assistance was used for code, silver-label preparation, analysis, and manuscript editing and is explicitly disclosed; it was not a human participant or independent assessor.

VIII. AVAILABILITY AND CONCLUSION

The sanitized release contains code, permitted metadata, sanitized experimental records, redistributable configurations, code-native figure sources, manifests, checksums, and reproduction instructions. It excludes PDFs, the live SQLite database, derived-text caches, interface screenshots, private

TABLE V
FULL-CORPUS PERFORMANCE ON DOCUMENTED LOCAL HARDWARE. TIMES ARE SECONDS; RSS IS MiB.

Stage	Units	Cold med.	Cold p95	Warm med.	Warm p95	Peak RSS	Fail.
Discovery (network)	59	3.240	3.917	0.970	2.833	65.2	0
PDF extraction	96	15.466	15.917	12.874	14.993	112.6	0
Chunking	96	10.404	10.773	8.280	9.853	62.5	0
Dense indexing	719	145.005	145.020	66.169	134.066	751.5	0
Keyword retrieval	6	9.809	10.126	8.091	9.526	111.8	0
Dense retrieval	6	84.487	86.265	12.215	78.342	563.5	0
Hybrid retrieval	6	93.355	93.616	20.326	86.385	568.9	0
Extractive answer	3	83.355	83.736	9.842	75.800	571.9	0
Recommendation	3	83.604	85.675	10.482	76.668	562.6	0
ASGI endpoints	7	79.606	81.178	3.948	72.397	641.2	0
Frontend build	1	31.145	33.201	30.763	31.270	461.2	0
Rendered routes	8	12.496	12.807	12.463	12.631	458.2	0

full-text prompts, secrets, model caches, and backups; no restricted-content redistribution is claimed.

For the frozen corpus, the platform makes research-intelligence failures inspectable and separates empirical quality from interface completeness. Complete eligible indexing, dense MRR 0.939, and claim support 0.995 coexist with a negative hybrid result, answer-point coverage 0.136, citation correctness 0.625, failed abstention, and an uncertain recommendation difference. It is therefore an evaluated research artefact and recommendation prototype, not a validated academic advisor. Human usefulness, identity decisions, third-party rights, and venue compliance remain external validations rather than values to invent.

REFERENCES

- [1] C. L. Giles, K. D. Bollacker, and S. Lawrence, "CiteSeer: An automatic citation indexing system," in *Proceedings of the Third ACM Conference on Digital Libraries*. New York, NY, USA: ACM Press, 1998, pp. 89–98. [Online]. Available: <https://doi.org/10.1145/276675.276685>
- [2] J. Tang, J. Zhang, L. Yao, J. Li, L. Zhang, and Z. Su, "ArnetMiner: Extraction and mining of academic social networks," in *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York, NY, USA: ACM, 2008, pp. 990–998. [Online]. Available: <https://doi.org/10.1145/1401890.1402008>
- [3] W. Ammar, D. Groeneveld, C. Bhagavatula, I. Beltagy, M. Crawford, D. Downey, J. Dunkelberger, A. Elgohary, S. Feldman, V. Ha, R. Kinney, S. Kohlmeier, K. Lo, T. Murray, H.-H. Ooi, M. Peters, J. Power, S. Skjonsberg, L. L. Wang, C. Wilhelm, Z. Yuan, M. van Zuylen, and O. Etzioni, "Construction of the literature graph in Semantic Scholar," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 3 (Industry Papers)*. New Orleans, Louisiana: Association for Computational Linguistics, Jun. 2018, pp. 84–91. [Online]. Available: <https://aclanthology.org/N18-3011/>
- [4] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," *arXiv preprint arXiv:2205.01833*, 2022. [Online]. Available: <https://arxiv.org/abs/2205.01833>
- [5] J. Beel, B. Gipp, S. Langer, and C. Breiteringer, "Research-paper recommender systems: A literature survey," *International Journal on Digital Libraries*, vol. 17, no. 4, pp. 305–338, 2016, published online in 2015; retrieved from the 2016 journal volume. [Online]. Available: <https://doi.org/10.1007/s00799-015-0156-0>
- [6] K. Sugiyama and M.-Y. Kan, "Scholarly paper recommendation via user's recent research interests," in *Proceedings of the 10th Annual Joint Conference on Digital Libraries*. New York, NY, USA: ACM, 2010, pp. 29–38. [Online]. Available: <https://doi.org/10.1145/1816123.1816129>
- [7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 9459–9474. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/6b493230-Abstract.html>
- [8] T. Narine and P. Hosein, "Automating the collection, display, summarization and podcasting of academic research," in *2025 IEEE 9th Forum on Research and Technologies for Society and Industry (RTSI)*. Tunis, Tunisia: IEEE, Aug. 2025, pp. 35–40. [Online]. Available: <https://doi.org/10.1109/RTSI64020.2025.11212505>
- [9] K. Lo, L. L. Wang, M. Neumann, R. Kinney, and D. Weld, "S2ORC: The semantic scholar open research corpus," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, Jul. 2020, pp. 4969–4983. [Online]. Available: <https://aclanthology.org/2020.acl-main.447/>
- [10] K. Balog, L. Azzopardi, and M. de Rijke, "Formal models for expert finding in enterprise corpora," in *Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. New York, NY, USA: ACM, 2006, pp. 43–50. [Online]. Available: <https://doi.org/10.1145/1148170.1148181>
- [11] —, "A language modeling framework for expert finding," *Information Processing & Management*, vol. 45, no. 1, pp. 1–19, 2009. [Online]. Available: <https://doi.org/10.1016/j.ipm.2008.06.003>
- [12] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 4, no. 1–2, pp. 1–174, 2009. [Online]. Available: <https://doi.org/10.1561/15000000019>
- [13] K. Weinberger, A. Dasgupta, J. Langford, A. Smola, and J. Attenberg, "Feature hashing for large scale multitask learning," in *Proceedings of the 26th Annual International Conference on Machine Learning*. New York, NY, USA: ACM, 2009, pp. 1113–1120. [Online]. Available: <https://doi.org/10.1145/1553374.1553516>
- [14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992. [Online]. Available: <https://aclanthology.org/D19-1410/>
- [15] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Online: Association for Computational Linguistics, 2020, pp. 6769–6781. [Online]. Available: <https://aclanthology.org/2020.emnlp-main.550/>
- [16] A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. Weld, "SPECTER: Document-level representation learning using citation-informed transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, 2020, pp. 2270–2282. [Online]. Available: <https://aclanthology.org/2020.acl-main.207/>
- [17] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021. [Online]. Available: <https://openreview.net/forum?id=wCu6T5xFjeJ>
- [18] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," in *Proceedings of the*

16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume. Online: Association for Computational Linguistics, 2021, pp. 874–880. [Online]. Available: <https://aclanthology.org/2021.eacl-main.74/>

- [19] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “SELF-RAG: Learning to retrieve, generate, and critique through self-reflection,” in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=hSyW5go0v8>
- [20] S. Es, J. James, L. E. Anke, and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*. St. Julians, Malta: Association for Computational Linguistics, 2024, pp. 150–158. [Online]. Available: <https://aclanthology.org/2024.eacl-demo.16/>
- [21] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, “ARES: An automated evaluation framework for retrieval-augmented generation systems,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Mexico City, Mexico: Association for Computational Linguistics, 2024, pp. 338–354. [Online]. Available: <https://aclanthology.org/2024.naacl-long.20/>
- [22] S. Min, K. Krishna, X. Lyu, M. Lewis, W. tau Yih, P. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi, “FactScore: Fine-grained atomic evaluation of factual precision in long-form text generation,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Singapore: Association for Computational Linguistics, 2023, pp. 12 076–12 100. [Online]. Available: <https://aclanthology.org/2023.emnlp-main.741/>
- [23] D. M. Blei, A. Y. Ng, and M. I. Jordan, “Latent dirichlet allocation,” *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003. [Online]. Available: <https://www.jmlr.org/papers/v3/blei03a.html>
- [24] M. Grootendorst, “BERTopic: Neural topic modeling with a class-based TF-IDF procedure,” *arXiv preprint arXiv:2203.05794*, 2022. [Online]. Available: <https://arxiv.org/abs/2203.05794>
- [25] C. Doogan and W. Buntine, “Topic model or topic twaddle? re-evaluating semantic interpretability measures,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, 2021, pp. 3824–3848. [Online]. Available: <https://aclanthology.org/2021.naacl-main.300/>
- [26] S. Amershi, M. Cakmak, W. B. Knox, and T. Kulesza, “Power to the people: The role of humans in interactive machine learning,” *AI Magazine*, vol. 35, no. 4, pp. 105–120, 2014. [Online]. Available: <https://doi.org/10.1609/aimag.v35i4.2513>
- [27] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, and K. Crawford, “Datasheets for datasets,” *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021. [Online]. Available: <https://doi.org/10.1145/3458723>
- [28] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, “Model cards for model reporting,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency*. New York, NY, USA: ACM, 2019, pp. 220–229. [Online]. Available: <https://doi.org/10.1145/3287560.3287596>
- [29] E. Tabassi, “Artificial intelligence risk management framework (AI RMF 1.0),” National Institute of Standards and Technology, Gaithersburg, MD, USA, Tech. Rep. NIST AI 100-1, Jan. 2023. [Online]. Available: <https://doi.org/10.6028/NIST.AI.100-1>